

Executive Summary

Galaxies change shape and structure over billions of years. Astronomers use a quantity called redshift to measure how far away a galaxy is, and how long ago its light was sent out. Knowing the redshift of a galaxy tells us when we are looking at it in cosmic history. Measuring redshift accurately usually needs spectroscopy, which is slow and expensive. Photometric redshift, based on images alone, is much faster but less accurate. Large future surveys, such as LSST and Euclid, will image billions of galaxies photometrically. Even where spectroscopic capability exists, it covers a small, selection-biased fraction of them. As a result, it is important to find better ways to connect galaxy images to redshift.

This project asks two questions. On one hand, given a target redshift, can we generate realistic galaxy images whose morphological properties reflect the evolutionary features expected at that cosmic time? On the other hand, given galaxy images, including real observed galaxies and synthetically generated galaxies, can we accurately recover their redshift and quantitative structural parameters? Together, these two directions form a bidirectional bridge between galaxy imaging and redshift.

The project builds on Lizarraaga et al. (2024), in which the authors trained a type of generative model called a Denoising Diffusion Probabilistic Model (DDPM) to create galaxy images conditioned on redshift. We reproduced this model using the same public dataset, called GalaxiesML (Li et al. 2024b), which contains around 286,000 real galaxy images with known redshifts. Our reproduced model matched the shape measurements of the original paper's images to within 2%. This confirms that the reproduction was accurate and gives a solid base to build on.

We then extended the work in several ways. First, the original paper used a rough shortcut to estimate a shape parameter called the Sérsic index, which describes how concentrated a galaxy's light is. However, the calculation of this shortcut is not actually related to the Sérsic index, and it is better described as an ellipticity-based proxy. We replaced this shortcut with a proper fit to the galaxy's brightness profile, giving a more physically correct measurement. Second, we trained a new neural network, which we call MorphCNN, to predict four shape parameters directly from a galaxy image. This is useful because it can not only check whether our generated images are physically consistent, but, if the model performs well, it can also act as a fast, automatic alternative to extraction tools such as SEP, which are more computationally expensive and require more manual tuning. Third, we improved the training process for the diffusion model, adding techniques such as early stop and reducing the learning rate when progress slows down. We also introduce new analysis methods, such as per-bin residual analysis of CNN redshift predictions, colour evolution analysis of generated samples, and a systematic comparison of different morphological measurement approaches including the effect of redshift label perturbation σ on generation quality.

Our experiments showed that the diffusion model successfully reproduces the correct trends in galaxy shape, size, and colour as redshift increases. Furthermore, a separate neural network, CNN redshift predictor, trained only on real galaxies, was able to correctly read the redshift back out of the generated images. This is strong evidence that the model has learned real physical patterns.

We also found some clear limitations, which are just as valuable as the successes. First, when we tested MorphCNN on the generated images, its predictions were noticeably worse than on real images. We traced part of this problem to the generated galaxies being systematically fainter at their centres than real ones. In addition, we found that two different measurement tools, SExtractor and SEP, give different answers for the same galaxy. This mismatch can make a model look worse than it really is, simply because it is being checked against inconsistent labels. This is an important finding for anyone using different software tools to measure galaxy properties. Finally, the dataset itself contains far more low-redshift galaxies than high-redshift ones. This imbalance makes both the diffusion model and the prediction models less reliable for the most distant, and often most scientifically interesting, galaxies.

Overall, the key achievement of this project is a complete and tested pipeline that connects galaxy image generation with redshift and physical evaluation. Lizarraga et al. (2024) were the first to show that a diffusion model conditioned continuously on redshift can implicitly learn the correlation between galaxy morphology and cosmic time. It addressed a gap that no reliable methods for predicting redshift given photometric images previously. It also improved previous generative models, which conditioned on redshift only at discrete, binned values. Based on the reproducing Lizarraga et al. (2024), our project verified the reliability of the method. Furthermore, our extension checks, in detail, whether those generated images obey the physical rules seen in real galaxies.

There are 3 next steps for this project. First, the model should be trained with data that gives more weight to rare, high-redshift galaxies, to reduce bias. Second, the shape and colour information from all five imaging bands should be used together, instead of relying mainly on one band. Third, the diffusion model could be conditioned on both redshift and shape parameters at the same time, rather than redshift alone. This would allow scientists to generate galaxies with chosen properties, which could support new studies of how galaxies form and change.

Looking ahead, this work points to two useful applications. First, generative models like this one could help create large, realistic datasets for future surveys, cutting down the cost of building datasets from real telescope time. Second, fast neural network tools like MorphCNN could help test the physical correctness of generated images. If such a model performs well on generated images, it also offers a promising route to improving on extraction tools such as SEP, making measurement faster and avoiding the errors introduced by different tool configurations. Together,

these tools could help astronomers make better use of the enormous volume of image data expected from the next generation of sky surveys.

References

Lizarraga A., Jiang E. H., Nowack J., Li Y. Q., Wu Y. N., Boscoe B., Do T., 2024, arXiv:2411.18440

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